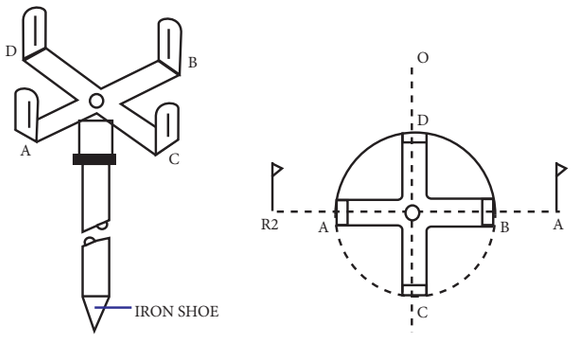
on each of the four sides. These are used for setting out right angles.



Open cross staff



French cross staff

Activity 1

Collect pictures of various chain surveying instruments and prepare an album.

9. Optical Square

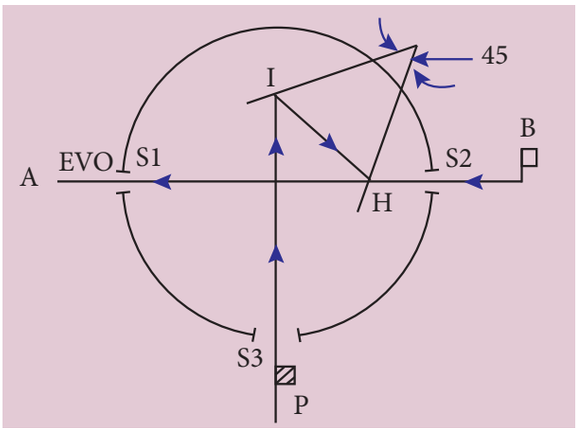
Optical square is an instrument used to construct perpendicular to a chain line.

It is circular in shape of 50 mm radius and 125 mm height. It has a sliding lid which cover the instrument when it is not in use.

As shown in figure, there are two reflecting mirrors H and I fixed at an angle of 45^0 to each other. The mirror H is half silvered and the mirror I is completely silvered. The three slits in the instrument are S_1 for eye sight S_2 is horizontal sight for ranging the station and S_3 is the index sight perpendicular to slits S_1 and S_2

The image of the ranging rod fixed at 'p' falls in the mirror I and gets reflected to the silvered portion of mirror H and reaches eye of the observer.

The ranging rod B can be viewed by the observer directly through the unsilvered portion of the mirror H. The ranging rod at 'P' is adjusted until the images of these two ranging rods coincide. Now the perpendicular is set for the chain line.



Optical Square

3.2.5 Ranging

The process of establishing intermediate points on a straight line between terminal points is known as ranging.

If the line is short (or) end of the station is clearly visible, it is easy to put the chain in the true alignment. But, if it is long or the end station is not clearly visible, it is necessary to place intermediate ranging rods to maintain the direction. It may be done by eye (or) instrumentally by using a line ranger.

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